

COURSE DESCRIPTION			
Program	Architecture		
Course Title	Architectural Design & Drafting I		
Course Code	2188	Semester	II
Type of Course	Studio	Contact Hours	75
Teaching Scheme (L:T:P/D:R)	0:1:4:0	Credits	3
CIE Marks	60	ESE Marks	40

Introduction

Introduces students to the basic principles of simple building design while developing essential drawing and presentation skills using standard architectural conventions.

Course Objectives

1	To introduce students to the fundamentals of designing simple single-storey buildings while becoming familiar with basic building rules and regulations.
2	To develop basic proficiency in producing clear and accurate presentation drawings using standard architectural conventions.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basics of Drafting and design development	1181	Architecture Graphics	I
Orthographic projection	1181	Architecture Graphics	I
Rendering	1182	Fundamentals of Architecture	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Discuss and understand the data related to spatial and functional planning incorporating building rules and regulations	Understand
CO2	Prepare detailed architectural drawings of single-story single room buildings	Apply
CO3	Prepare detailed rendered architectural drawings of single-story buildings	Apply
CO4	Prepare detailed rendered architectural drawings of single-story buildings with perspective drawings and a 3D model.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	3	3	3	2	3	1	2	1	-	3
CO2	3	3	3	3	3	3	2	2	2	2	3
CO3	3	3	3	3	3	3	2	2	2	2	3
CO4	3	3	3	3	3	3	2	2	2	2	3
Total	11	12	12	12	11	12	7	8	7	6	12
Correlation Level	2.67	3	3	3	2.67	3	1.75	2	1.75	2	3

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
L	T	P	R				CIE	ESE	Total	CIE	ESE	Total
0	1	4	0	3	75	3	0	0	0	60	40	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			T	D
Module I	Data Collection	Design Process, Data collection in the design process - Human anthropometry, Data collection on anthropometric standards, spatial standards, Anthropometry exercise - Case study methods and analysis, case study of a single room building	2	8
Module II	Single-storey building- Watchman Cabin	Design problem - Data collection on anthropometric requirements and area standards - Case study of similar projects - Basic sketch scheme and bubble diagram for functional zoning - Preparation of scaled plans, elevations and sections	2	8
Module II	Bus shelter	Design problem - Data collection on anthropometric requirements and area standards for waiting areas - Case study - Basic sketch scheme and bubble diagram for functional zoning - Preparation of scaled drawings: Plans, elevations and sections	2	8
Module III	Single-storey building with multiple rooms- Design a Cafeteria (below 75 sq m)	Design problem - Data collection on area standards, circulation, service areas, and anthropometric standards - Case study and analysis - Preparation of basic sketch scheme and bubble diagram for functional zoning - Preparation of detailed presentation drawings: Plans, elevations and sections	3	18
Module IV	Single-storey residential building with multiple rooms- Weekend home (below 100 sq m)	Design problem - Data collection on anthropometric requirements and area standards for waiting areas - Case study and analysis - Basic sketch scheme and bubble diagram for functional zoning - Detailed presentation drawings: Plans, elevations and sections, two-point perspective of the building exterior views - Scaled model of design	4	20

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (T/S)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Introduction to design	Discuss the basics of design process- Understand anthropometric data and its use in design	L	CO1	Understand	1
Introduction to design	Summarize data from Time Savers, Architects' Handbook, and KMBR.	P	CO1	Understand	4
Introduction to case study	Discuss the methods of case study and its analysis	L	CO1	Understand	1
Introduction to case study	Discuss the importance of case studies in design development	P	CO1	Understand	4
Module II					
Single-storey building- Watchman Cabin	Discuss the details of the project and data collection	L	CO2	Understand	2
Single-storey building- Watchman Cabin	Summarize collected data and prepare sketch scheme	P	CO2	Apply	4
Single-storey building- Watchman Cabin	Prepare a detailed Plan, sections and elevations.	P	CO2	Apply	4
Bus shelter	Discuss the details of the project and data collection	L	CO2	Understand	2
Bus shelter	Summarize collected data and prepare sketch scheme	P	CO2	Apply	4
Bus shelter	Prepare a detailed Plan, sections and elevations.	P	CO2	Apply	4
Module III					
Single-storey building with multiple rooms- Design a Cafeteria (below 75 sq m)	Discuss the details of the project and data collection	L	CO3	Understand	3

Single-storey building with multiple rooms- Design a Cafeteria (below 75 sq m)	Conduct relevant case studies to understand various design elements—do comparative analysis and infer from the studies.	P	CO3	Apply	4
Single-storey building with multiple rooms- Design a Cafeteria (below 75 sq m)	Prepare conceptual sketches, bubble diagrams, spatial zoning and form, preliminary plan	P	CO3	Apply	4
Single-storey building with multiple rooms- Design a Cafeteria (below 75 sq m)	Prepare detailed Plan, sections and elevations	P	CO3	Apply	4
Single-storey building with multiple rooms- Design a Cafeteria (below 75 sq m)	Prepare presentation drawings of the design with rendering techniques (explore different types of color mediums, rendering styles etc).	P	CO3	Apply	6
Module IV					
Single-storey building with multiple rooms- Weekend home (below 100 sq m)	Discuss the details of the project and data collection	L	CO4	Understand	4
Single-storey building with multiple rooms- Weekend home (below 100 sq m)	Conduct relevant case studies to understand various design elements—do comparative analysis and infer from the studies.	P	CO4	Apply	4
Single-storey building with multiple rooms- Weekend home (below 100 sq m)	Prepare conceptual sketches and form a preliminary plan	P	CO4	Apply	4
Single-storey building with multiple rooms- Weekend home (below 100 sq m)	Prepare detailed Plan, sections and elevations	P	CO4	Apply	4
Single-storey building with multiple rooms- Weekend home (below 100 sq m)	Construct two-point perspective views of the building and render the drawing.	P	CO4	Apply	2
Single-storey building with multiple rooms- Weekend home (below 100 sq m)	Prepare a model of the building. (explore various materials such as forex sheets /cardboard/mount board/cards sheets, etc)	P	CO4	Apply	6

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Measure the anthropometric dimensions of a person in a sitting posture and prepare a neatly labelled anthropometric sketch.	3
2	Analyse the height of your table with respect to the anthropometric data and standards	3
3	Study and list out the basic terminologies mentioned in KMBR	3
4	Write an analysis of an already conducted case study	3
Module II		
5	Watch videos on international examples of bus shelters and list out the possible construction materials and structural typologies that can be adapted in design	3
6	Analyse the case study of the bus shelter with respect to anthropometric standards, climatic conditions, and identify issues in the existing design	3
7	Analyse and list out the merits and demerits of the design of the watchman's cabin in your college.	3
8	Prepare a bubble diagram for the watchman's cabin	3
Module III		
9	Measure and sketch the dining table and chairs in your house. (Understand spatial requirement)	3
10	Watch videos on international cafeteria case studies and list out the ideas that can be incorporated in your design	3
11	Sketch a bubble diagram of the proposed cafeteria	3
12	Explore different sheet presentation media and techniques using online resources	3
Module IV		
13	Study how people use weekend homes through short documentaries, YouTube house tours, or case-study videos. Sketch and denote activity-space and design criteria that can be adapted for weekend home design	3
14	Explore materials suitable for weekend home design and make a material album.	3
15	Take a realistic photograph of created model and render it using AI	3

Total Self-Learning hours	45
---------------------------	----

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Neufert Architect's Data	Ernst Neufert	Wiley
2	Timesaver standards for building types	Joseph De Chiara and John Callendar	McGRAW-HILL
3	Kerala Municipality Building Rules	Town Planning Department Kerala	

REFERENCE			
Sl. No.	Title	Author	Publication
1	Rendering with pen and ink	Robert W Gill	Thames and Hudson
2	Design Drawing	Fracis DK Ching	Wiley
3	Human Dimension and Interior Space	Julius Panero, Martin Zelnik	Whitney Library of Design

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.kaarwan.com/blog/architecture/incorporating-anthropometric-data-into-building-codes-standards?id=840&	i
2	https://guides.library.ucla.edu/architecture/manualsandstandards?	II & III

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	NIL	NIL	NIL

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	NIL	NIL	NIL

Assessment Methodology

Mode	Attendance	CA1		CA2		CA3		CA4		CA5	CA6	ESE
		Continuous evaluation (Dwg sheets-Mod 1)	Self-learning activities (Module 1)	Continuous evaluation (Dwg sheets-Mod 2)	Self-learning activities (Module 2)	Continuous evaluation (Dwg sheets-Mod 3)	Self-learning activities (Module 3)	Continuous evaluation (Dwg sheets-Mod 4)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written exam (Drawing)
Duration										2 hours	2 hours	3 hours
Exam mark		30	10	30	10	30	10	30	10	50	50	40
Converted to		40		40		40		40		15	15	
Marks	5	40								15		40
Total	60											40
Tentative schedule	End of semester	By 4th week		By 7th week		By 11th week		By 14th week		8th week	15th week	End of semester
Tentative schedule	End of semester	By 4th week	By 4th week	By 7th week	By 7th week	By 11th week	By 1th week	By 14th week	By 14th week	8th week	15th week	End of semester

Scheme of Evaluation - External assessment

Sl. No.	Category	Description	Marks	Time
1	Drawing exam	Drafting skills, technical knowledge, and quality of drawing will be evaluated through a drawing exam.	20	3 hrs
2	Design Jury (Portfolio)	Design quality, design development, functional planning, rendering, and presentation skills will be evaluated through a design jury consisting of an expert panel.	20	

External exam question paper format - Drawing (20 marks)

Module	Topic	Part A	Blooms Taxonomy	Total
2,3,4	Prepare detailed drawings of any one project	1	Apply	20
	Total	1	-	20

Scheme of Evaluation (External Exam) - Drawing (20 marks)

Sl. No.	Major topics	Criteria	Description	Marks
Part A	Detailed drawings of the project	Design	Quality of design and correctness of functional areas	10
		Drafting and detailing	Drafting correctness and quality, dimensioning, annotation, detailing	10
Total				20

Scheme of Evaluation (External Exam) - Design Jury (20 marks)

Major topics	Criteria	Description	Marks
All projects of the semester	Design	Quality of design, concept/design development, and correctness of functional areas and correctness of functional areas	8
	Presentation	Quality of sheets and portfolio	6
	Drafting and detailing	Drafting correctness and quality, dimensioning, annotations, detailing	6
Total			20

SITTTR